

GPUPilot: Universal AI GPU Performance Engineer

Hackathon Project Presentation & Architectural Walkthrough

- Autonomous performance copilot connecting to NVIDIA, AMD, Intel & WebGL accelerators
- Real-time 6-rule bottleneck detection with live confidence scoring
- Automated PyTorch code remediation (AMP FP16, activation checkpointing, pin memory)
- Standardized 0–1000 composite stress benchmarking with thermal delta tracking
- Dual-Engine AI Workload Explainer (Google Gemini 1.5 Flash + 100% offline fallback)
- Autonomous silicon power & performance tuning profiles with dry-run safety

The Problem: Why GPU Optimization is Opaque

Critical hardware bottlenecks that stall models and degrade performance

- Silent Thermal Throttling: Silicon safety circuits drop clocks by up to 60% without warning
- CUDA Out-Of-Memory (OOM): Catastrophic crashes on consumer 4GB to 8GB VRAM accelerators
- Host CPU Starvation: GPU sits idle at 20% while single-thread Python DataLoaders spike to 100%
- Memory Bandwidth Chokepoints: Unpinned host memory transfers and PCIe bus stalls
- Tooling Gap: Existing tools like nvidia-smi show raw percentages, but zero actionable code fixes

The Solution: GPUPilot Architecture

An autonomous AI performance engineer running locally and in the browser

- 6-Rule Bottleneck Engine: Prioritized diagnostics (Thermal, VRAM OOM, Compute, CPU Starvation)
- Automated Remediation Engine: Verified, copy-paste PyTorch prescriptions with impact ratings
- Multi-Stream Rolling Telemetry: Live charts toggling Core View vs CPU and Power View
- GPU Benchmarking Suite: Matrix multiplication and memory stress with 0–1000 composite scoring
- Autonomous Tuner: Power Saver, Balanced, and Maximum Performance execution profiles

Modular Hardware Abstraction Layer (HAL)

Cross-vendor compatibility across NVIDIA, AMD, Intel, and WebGL

- NVIDIA Provider: Production NVML (pynvml) for temperature, power, clocks, and VRAM
- AMD Provider: amdsmi SDK integration with safe per-sensor fallback
- Intel Provider: Windows DXGI engine performance counters and WMI system integration
- WebGL WebEngine: In-browser simulation canvas for zero-install evaluation on Vercel
- Comprehensive Test Suite: 42 of 42 automated backend unit tests passing cleanly

Physical Hardware Validation & Case Study

Verified on NVIDIA GeForce RTX 3050 4GB Laptop GPU + Intel Core i5

- Stress Scenario: 30,000 WebGL entity simulation on physical ASUS VivoBook laptop
- Observed Vitals: Intel CPU reached 92°C while dedicated RTX 3050 remained at 62°C (34% load)
- Real-Time Diagnosis: Flagged CPU_STARVATION with 94% confidence score
- Automated Fix: Advised offloading transforms to compute shaders and setting num_workers to 4
- Result: Eliminated draw-call bottlenecks and balanced the shared chassis thermal envelope

Dual-Engine AI Workload Explainer (Phase 9)

Conversational LLM reasoning grounded in live telemetry + 100% offline fallback

- □ Google Gemini 1.5 Flash Mode: Connect via free API key for full conversational depth
- Live Telemetry Grounding: Real-time metrics automatically injected into systems-engineering prompts
- Zero VRAM Footprint: Cloud LLM execution leaves 100% of local GPU memory free for user models
- □ Heuristic Offline Engine: Answers 12 distinct GPU categories in under 5 milliseconds
- Zero Friction for Judges: Operates completely offline or on Vercel without requiring an API key

GPU Benchmarking Suite & Autonomous Tuner

Stress testing resilience and 1-click silicon power management

- 0–1000 Normalized Score: Weighted by Compute (40%), Thermal Resilience (35%), Memory (25%)
- Live Telemetry Delta: Tracks peak temperature rises and clock downclock margins under load
- Power Saver (Efficiency) Profile: Caps power envelope by 20%, drops temps by 10°C, acoustic quiet
- Balanced Profile: Dynamic voltage-frequency scaling suitable for daily gaming and rendering
- Maximum Performance Profile: Maximizes batch queue depth and compute pipeline allocation

Production Ready: 100% Phases Complete

Live across physical hardware and global cloud deployment

- 10 of 10 Roadmap Phases: Fully implemented, integrated, and verified
- 42 of 42 Unit Tests Passing: Rock-solid test coverage across backend modules
- Global Deployment: Live on Vercel at <https://gpu-pilot.vercel.app/>
- Open Source Codebase: Tracked on GitHub at [Harshityadav9838/GPUPilot](https://github.com/Harshityadav9838/GPUPilot)
- Offline Archive: Standalone GPUPilot.zip package ready for hackathon evaluation